

CHIRANJIVI BHATTARAI, PHD

800 Ralston street, Reno, NV, USA | 775 - 200 - 4568 | chiranjivibh@gmail.com | [chiranjivi-bhattarai-029a745a](https://orcid.org/0000-0002-9a74-5a)

SUMMARY

- Interdisciplinary scientist with skills and experience in organic analytical chemistry, aerosols, and air quality
- Led collaborative projects, resulting in 12 peer-reviewed publications
- Deep understanding of data analysis and visualization
- Self-motivated, problem-solving, and collaborative scientist with excellent communication skills
- Looking to contribute to computational methods to push forward atmospheric chemistry modeling and toxicity to get an early assessment of the air quality.

TECHNICAL SKILLS

- **Analytical skills:** Gas chromatography-mass spectroscopy (GC-MS), Liquid chromatography-VV (LC/MS), (UV spectrometry, TD/GC/MS, HPLC/UV, UPLC/MS), and accelerated solvent extractor (ASE).
- **Online instruments:** Scanning mobility particles sizer (SMPS), tandem thermodenuder SMPS (TD-SMPS), photo-acoustic soot spectrometer (PASS), Single Particle soot photometer (SP₂), and CO/CO₂/NO_x/Ozone analyzers
- **Computational:** Programming languages (Python, GitHub, and Matlab)

RESEARCH EXPERIENCE

Desert Research Institute

Research Scientist

February 2022 to present

The core research focused on thermodynamic proprieties of the laboratory and ambient aerosols using tandem differential mobility analyzer (TDMA) and gas/particles phase chemical specification of the emissions.

- CRC survey: The CRC fuel survey data was employed to calculate the particulate matter index (PMI), utilizing various analytical techniques such as gas chromatography (GC) and distillation. Data harmonization across these methods was essential for consistent PMI results. This was followed by comprehensive statistical analyses for effective data interpretation.
- NIH biomass burning: Lead a project to design and conduct a laboratory burning experiment. Chemical specification was carried. Analyzed the samples using GC-MS. In addition, I also performed volatility tandem differential mobility analyzer experiments to study thermodynamic properties of emission.
- Experienced in sample analysis for chemical specifications of organic compounds using GC - MS, LC - MS and LC- VU/vis to understand emission factors from different sources.
- Experience with mentoring graduate and undergraduate students (mentored three graduate and one undergraduate) and colleagues (two staffs) for general lab operations. Also, instructed/trained students for ambient/laboratory air quality sample collection, extraction, and analysis of the samples.
- Manage general laboratory administrative process for keeping the track of the laboratory supplies and regular routine analytical instrument maintenance (GC - MS, LC).
- Prepare SOP for the instruments, protocols for lab operation, and lab safety instructions

Desert Research Institute

Graduate Research Assistant

August 2014 to December 2021

- NSF biomass burning: Design experimental setup, Oxidation flow reactor calibration, conduction of experiments, sample collection, extraction, quantification and analysis
- E-Cigarette project: Conducting experiment, sampling, extraction, quantification and analysis of the emissions
- NIH biomass burning emissions: data collection, quantification and analysis of the samples

EDUCATION

- PhD, Atmospheric Sciences, University of Nevada, Reno, 2021
- Masters, Atmospheric Sciences, University of Nevada, Reno, 2018
- Masters, Physics, Tribhuvan University, Kathmandu, Nepal, 2011

TEACHING AND MENTORING EXPERIENCE

- 2022 - Mentored 4 graduate and 2 undergraduate students in their day-to-day lab activities and project executions
- 2018 - 2021 Graduate Teaching Assistant in Physics, taught undergraduate level physics classes:- Physics 180L, Physics 152L, and instructed general physics 180.

AWARDS

- GSA's Fall 2020 Outstanding Graduate Student award
- GSA's Spring 2020 Outstanding International Student award
- GSA International Student Scholarship
- Graduate student travel grants

CONFERENCE PRESENTATIONS

Selected conference presentations

- Bhattacharai, Chiranjivi** and Andrey Y Khlystov (Oct. 2016). "An Algorithm to Derive particle Size Changes from Full Size Distribution Measurements". In: *2016 AAAR Annual Conference Abstracts*. American Association for Aerosol Research. Portland, OR: AAAR.
- Bhattacharai, Chiranjivi**, Vera Samburova, and Andrey Y Khlystov (Oct. 2016). "Characterization of Aerodyne Potential Aerosol Mass Oxidative Flow Reactor". In: *2016 AAAR Annual Conference Abstracts*. American Association for Aerosol Research. Portland, OR: AAAR.
- Bhattacharai, Chiranjivi**, Deep Sengupta, Michealene laukea-Lum, Vera Samburova, Adam C Watts, Hans Moosmüller, and Andrey Y Khlystov (Oct. 2017). "Physical and Chemical Characterization of Fresh and Aged Emissions from Open Combustion of Biomass Fuels". In: *2017 AAAR Annual Conference Abstracts*. Ed. by AAAR. American Association for Aerosol Research. Raleigh, NC: AAAR.
- Bhattacharai, Chiranjivi**, Deep Sengupta, Vera Samburova, Adam C Watts, Hans Moosmüller, and Andrey Y Khlystov (May 2017). "Physical and Chemical Characterization of Fresh and Aged Emissions from Open Combustion of Biomass Fuels". In: Desert Research Institute Graduate student poster competition. Reno, NV: DRI.
- Bhattacharai, Chiranjivi** and Andrey Y Khlystov (Oct. 2019). "Volatility-Based Measurements of Aerosol Mixing State at an Urban Background Site in the Western United States". In: *2019 AAAR Annual Conference Abstracts*. American Association for Aerosol Research. Portland, OR: AAAR.
- (Oct. 2020). "Improvement on Differential Mobility Analyzer Method for Estimation of the Enthalpy of Vaporization Using an Algorithm to Derive Particle size Changes from Full-Size Distribution". In: *2020 AAAR Annual Conference Abstracts (Virtual)*. American Association for Aerosol Research. Raleigh, NC: AAAR.

PUBLICATIONS

Selected Peer-reviewed Journal Articles

- Bhattacharai, Chiranjivi**, Vera Samburova, Deep Sengupta, Michealene laukea-Lum, Adam C Watts, Hans Moosmüller, and Andrey Y Khlystov (2018). "Physical and chemical characterization of aerosol in fresh and aged emissions from open combustion of biomass fuels". In: *Aerosol Science and Technology* 52.11, pp. 1266–1282. DOI: [doi:10.1080/02786826.2018.1498585](https://doi.org/10.1080/02786826.2018.1498585).
- Samburova, Vera, **Chiranjivi Bhattacharai**, Matthew Strickland, Lyndsey Darrow, Jeff Angermann, Yeongkwon Son, and Andrey Khlystov (2018). "Aldehydes in Exhaled Breath during E-Cigarette Vaping: Pilot Study Results". In: *Toxics* 6.3, p. 46.
- Sengupta, Deep, Vera Samburova, **Chiranjivi Bhattacharai**, Elena Kirillova, Lynn Mazzoleni, Michealene laukea-Lum, Adam Watts, Hans Moosmüller, and Andrey Khlystov (2018). "Light absorption by polar and non-polar aerosol compounds from laboratory biomass combustion". In: *Atmospheric Chemistry and Physics* 18.15, pp. 10849–10867.
- Son, Yeongkwon, **Chiranjivi Bhattacharai**, Vera Samburova, and Andrey Khlystov (2020). "Carbonyls and carbon monoxide emissions from electronic cigarettes affected by device type and use patterns". In: *International journal of environmental research and public health* 17.8, p. 2767.
- Bhattacharai, Chiranjivi** and Andrey Khlystov (2021). "Derivation of particle-size changes from polydisperse size distribution measurements: numerical and experimental verification". In: *Aerosol Science and Engineering*, pp. 1–9.
- Axelrod, Kevin, **Chiranjivi Bhattacharai**, Palina Bahdanovich, Vera Samburova, and Andrey Y Khlystov (2023). "The volatility of pollen extracts and their main constituents in aerosolized form via the integrated volume method (IVM) and the volatility basis set (VBS)". In: *Aerosol Science and Technology* just-accepted, pp. 1–19. DOI: <https://doi.org/10.1080/02786826.2023.2265954>.
- Sengupta, Deep, Vera Samburova, **Chiranjivi Bhattacharai**, Hans Moosmüller, and Andrey Khlystov (2023). "Emission factors for polycyclic aromatic hydrocarbons from laboratory biomass-burning and their chemical transformations during aging in an oxidation flow reactor". In: *Science of The Total Environment* 870, p. 161857.

For more details, see [google scholar](#)

OTHER SKILLS

Software Python, Matlab, Microsoft Word, Excel, and PowerPoint, LaTeX, Gaussian

Languages English: professional proficiency. Nepali: native.